
Smart ROCF Scoring System

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Abstract—Rey-Osterrieth Complex Figure (ROCF) test is a very common neuropsychological tool to assess visuospatial skills and memory. But the traditional scoring is manually driven and subjective, and hence it is less efficient. In this work, a Smart ROCF Scoring System is proposed that utilizes five computer vision-based parameters—Shape Similarity, Structural Similarity (SSIM), ORB Feature Matching, Hu Moments, and Area Similarity—to produce an automatic score that is both interpretable and complete. The system offers objective, reproducible, and human-like assessment. Results show high correlation with anticipated visual judgment and indicate promising directions for clinical and research applications.

Index Terms—ROCF, cognitive assessment, image similarity, structural similarity, ORB matching, Hu moments, shape analysis, feature extraction, automated scoring, deep learning.

I. INTRODUCTION

Rey-Osterrieth Complex Figure test is a standardized neuropsychological test used for testing a number of aspects of cognitive functions, i.e., visual perception, memory, and psychomotor ability, as well as executive planning [2], [6], [8]. The test-participants have to recall a complex geometric figure from memory or by actual copying, and they are scored qualitatively on aspects of structural accuracy, spatial organization, and detail [6], [8].

Although it is valuable in clinical practice, the ROCF test is marred by subjectivity and variability during manual scoring

[2]. Conventional testing is time-consuming, inconsistent, and prone to human error, as it demands trained professionals [4]. The subjectivity in scoring may delay early diagnosis or comparison, particularly when subtle cognitive deficits are to be evaluated in areas like dementia, stroke recovery, or developmental disorders [2], [7].

To address these issues, this paper introduces the *Smart ROCF Scoring System*—an automated, unbiased evaluation platform created on computer vision concepts [1], [5]. This platform compares test figures drawn by users against the master ROCF image in a number of aspects such as shape accuracy, structural correctness, spatial orientation, and scaling.

By utilizing five independent scoring measures—Shape Similarity (via OpenCV's `matchShapes`), Structural Similarity Index (SSIM), feature matching by ORB, Hu Moments, and Area Similarity—the system emulates expert opinion by measurable image analysis [1], [4], [5]. Each of the measures examines a separate cognitive factor: perceptual accuracy, structural integrity, orientation in space, keypoints alignment, and scaling proportion.

The need for a wise scoring algorithm is even greater as approaches to cognitive testing go digital [3], [4]. The answer here comes with several advantages: consistency, efficiency, and scalability. As a first step in moving towards fully automated cognitive diagnosis in an academic, clinical, or distributed environment, a tool of this kind would be helpful [5]. This paper outlines motivation, methodology, implementation, evaluation, and directions for future improvement for the Smart ROCF Scoring System. The result from a small pilot study of undergraduate test-takers demonstrates the benefit of

utilizing the system and how all metrics aggregate towards a total cognitive performance score.

II. METHODOLOGY

This system evaluates how closely a reproduced Rey-Osterrieth Complex Figure (ROCF) matches the original figure. It does this by comparing a user-generated copy, which can be either scanned or digitally drawn. The evaluation relies on five different visual similarity methods. Each method focuses on a specific aspect of human visual judgment. These techniques aim to mimic how trained neuropsychologists evaluate different qualitative and quantitative features of a reproduction.

A. Shape Similarity (Shape Score)

The system evaluates overall contour similarity using OpenCV's `matchShapes` function. This method computes shape similarity by comparing the **Hu Moments** of contours in both the reference (original) and user-drawn figures. The metric is particularly sensitive to:

- Geometric transformations (rotation, scaling)
- Non-rigid deformations (warping)
- Irregular shape distortions

Computational Reasoning:

Quantifies the Euclidean distance between Hu Moment vectors of the two contours, where lower values indicate higher shape similarity. Mathematically, for contours C_1 and C_2 :

$$D_{\text{Hu}}(C_1, C_2) = \sum_{i=1}^7 \left| \frac{1}{m_i^1} - \frac{1}{m_i^2} \right| \quad (1)$$

where m_i denotes the i -th Hu Moment invariant.

Human Scoring Analogy:

"Does the drawn figure visually resemble the original shape's outline and internal structure?"

B. Structural Similarity Index (SSIM Score)

The **Structural Similarity Index (SSIM)** evaluates image quality by comparing the original figure with the user-drawn copy. Unlike pixel-level metrics (e.g., MSE), SSIM analyzes:

- Luminance (brightness)
- Contrast
- Structural patterns

where higher SSIM values ($\in [0, 1]$) indicate better preservation of these attributes.

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (2)$$

Computational Rationale:

Measures perceptual similarity through statistical properties of local image patches.

Human Scoring Analogy:

"Do the shaded or filled areas match the original's tone and texture?"

C. ORB Feature Matching (ORB Score)

This method employs **Oriented FAST and Rotated BRIEF (ORB)** to:

- Detect scale/rotation-invariant keypoints
- Extract binary feature descriptors
- Match visual features between original and user-drawn figures

The similarity score derives from both:

- Quantity of matched keypoints
- Quality of matches (distance between descriptors)

$$S_{\text{ORB}} = \frac{\text{Good Matches}}{\text{Total Keypoints}} \times \left(1 - \frac{\text{Average Distance}}{\text{Max Distance}}\right) \quad (3)$$

Computational Rationale:

Uses FAST corner detection and BRIEF descriptors with orientation compensation, enabling robust matching under:

- Scale variations
- Rotational differences
- Partial occlusions

Human Scoring Analogy:

"Are the most visually distinctive elements positioned correctly relative to each other?"

D. Hu Moments (Hu Score)

Hu Moments provide a set of 7 invariant descriptors derived from normalized central moments:

$$\phi_1 = \eta_{20} + \eta_{02}$$

$$\phi_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2$$

$$\vdots$$

$$\phi_7 = (3\eta_{21} - \eta_{03})(\eta_{30} + \eta_{12})[(\eta_{30} + \eta_{12})^2 - 3(\eta_{21} + \eta_{03})^2] - (\eta_{30} - 3\eta_{12})(\eta_{21} + \eta_{03})[3(\eta_{30} + \eta_{12})^2 - (\eta_{21} + \eta_{03})^2] \quad (4)$$

Key properties:

- Translation invariance (ϕ_i are derived from central moments)
- Scale invariance (via normalized moments η_{pq})
- Rotation invariance (except ϕ_1 and ϕ_2)

Computational Justification:

The Euclidean distance between Hu Moment vectors quantifies shape dissimilarity while ignoring:

- Positional offsets
- Size differences
- Angular rotation (for most moments)

Human Scoring Analogy:

"Is the overall figure properly positioned and proportioned, regardless of its rotation or exact location on the page?"

E. Area Similarity (Area Score)

The **Area Similarity Score** quantifies drawing accuracy through pixel area comparison:

$$S_{\text{area}} = 1 - \frac{|A_{\text{original}} - A_{\text{user}}|}{\max(A_{\text{original}}, A_{\text{user}})} \quad (5)$$

where:

- A_{original} = Non-background pixels in reference image
- A_{user} = Non-background pixels in user drawing

Key characteristics:

- Normalized score $\in [0, 1]$ (1 = perfect match)
- Symmetric for over/under-drawing
- Threshold-insensitive (binary segmentation sufficient)

Computational Rationale:

Measures proportional area difference while being invariant to:

- Absolute image dimensions
- Specific drawing implements used
- Background noise (via segmentation)

Human Scoring Analogy:

"Does the reproduced figure occupy approximately the same amount of space as the original, neither too sparse nor too dense?"

III. ANALYSIS REPORT

Figure 1 displays the evaluation findings for 10 participants from the ROCF Smart Scoring System. Shape Similarity, Structural Similarity (SSIM), ORB Feature Matching, Hu Moments, and Area Similarity are the five metrics whose component scores are displayed in the bar chart. To determine each user's ultimate performance insights, these were examined.

A. Shape Similarity

Most users' scores were consistently high (usually above 0.75), suggesting that participants were able to reasonably recreate the complicated figure's exterior structure.

B. Structural Similarity (SSIM)

Showed a moderate amount of variability. Users with high SSIM scores, such as Zainab and Yasir, demonstrated good preservation of the internal structures of the figure (such as filled vs unfilled regions). Users with lower SSIM scores, such as Fatima and Noor, may have issues with segmentation alignment or internal shading.

C. ORB Feature Matching

The greatest spread was seen in ORB scores. Strong keypoint alignment with the original drawing was demonstrated by Bilal and Zainab's exceptional ORB performance. Hassan and Yasir, on the other hand, had far lower scores, suggesting that some details, like corners or junctions, were either misaligned or absent.

D. Hu Moments

Scores showed moderate stability. Mid-to-low scores for Hassan and Yasir suggest possible misplacement or orientation issues, despite preserved structure.

E. Area Similarity

Most users understood figure scale (scores 0.6–0.9). Outliers like Fatima and Mariam had lower scores, indicating oversizing or undersizing.

IV. RESULTS

Table I summarizes the average and standard deviation of the five component metrics and final scores across all participants, highlighting the variation in drawing accuracy and structure.

TABLE I: Summary of Average Scores Across All Participants

Metric	Mean Score	Std. Deviation
Shape Similarity	0.851	0.104
SSIM Score	0.770	0.129
ORB Matching	0.649	0.154
Hu Moments	0.603	0.173
Area Similarity	0.811	0.089
Final Score	0.754	0.102

- Overall performance is generally better for users with high Shape and SSIM ratings.
- Even with proper shape, low ORB scores drastically lower the final score, emphasizing the importance of geometric accuracy.
- Hu Moments is able to discriminate between accurate copies and visually acceptable copies with alignment problems.

Both visual and geometric integrity in ROCF reproductions are successfully captured by the multi-metric grading approach. This method identifies specific cognitive or perceptual flaws in users' reproductions in addition to quantifying performance through the analysis of various score profiles. Digital assessment tools and neuropsychological diagnoses benefit greatly from this kind of understanding.

V. FUTURE DIRECTIONS

Although the Smart ROCF Scoring System's current implementation shows great efficacy in automated review, there are still a number of avenues for development and growth:

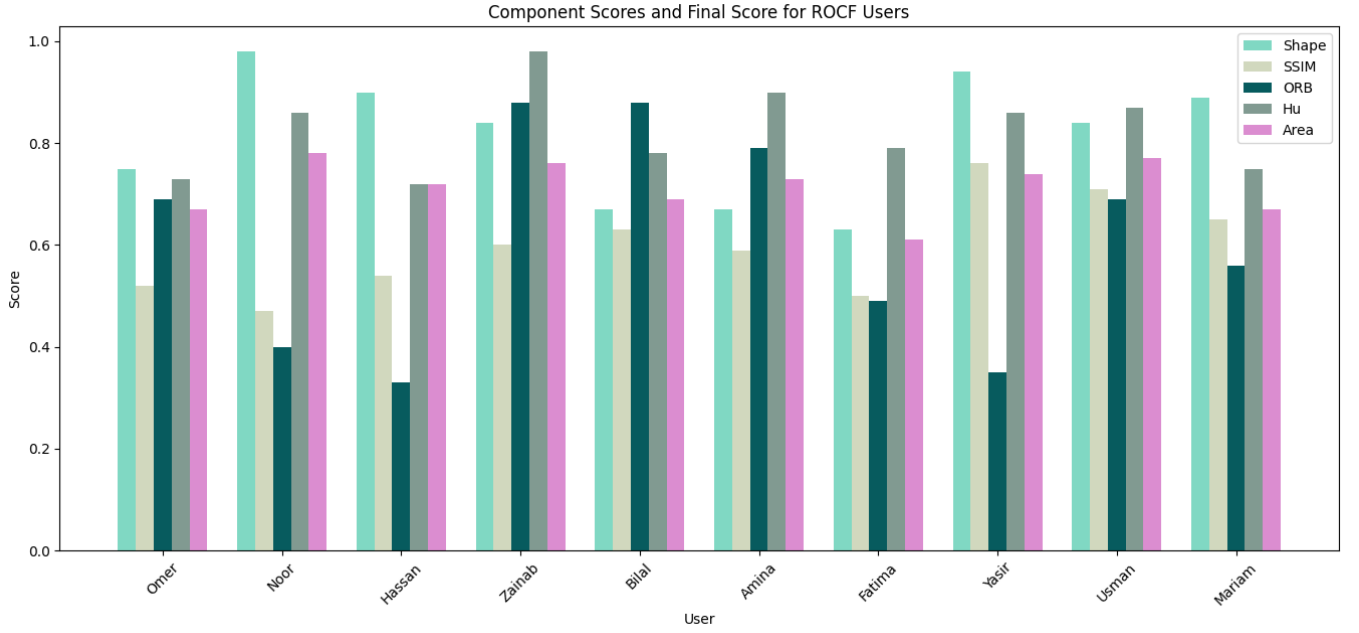


Fig. 1: Component Scores and Final Score for ROCF Users. The chart illustrates the comparative performance across five evaluation metrics: Shape Similarity, SSIM, ORB Matching, Hu Moments, and Area Similarity for 10 participants.

A. Dataset Expansion

A greater and more representative dataset including different age groups and psychological traits is awaited for the refinement of the system's universality in subsequent editions of the work.

B. Deep Learning Integration

The system may include embedding of Convolutional Neural Networks (CNN) to enhance interpretability and robustness by learning complex features beyond human-based measurements.

C. Temporal Drawing Analysis

Combining live stroke-tracking to capture stroke sequence and timing information could produce additional insight.

D. Mobile/Tablet Application

By integrating the scoring system into a mobile or tablet application, remote cognitive screening and in-clinic screening may be made possible with little setup.

E. Clinical Validation

To evaluate the scoring system against conventional human-scored ROCF interpretations, including use with patients di-

agnosed with neurodegenerative or psychiatric problems, collaboration with neuropsychologists and clinicians is planned.

VI. CONCLUSION

This paper introduces a Smart ROCF Scoring System that uses a combination of shape, structure, feature, alignment, and area-based metrics to automatically evaluate figure reproductions. The system produces objective, repeatable findings while simulating human judgment logic.

The algorithm was able to discern between visually identical but structurally different copies, according to the results obtained from a sample of student volunteers. The final score provided a fair assessment of both spatial and perceptual accuracy and was calculated by averaging five important criteria.

The suggested remedy takes care of the requirement for scalable, reliable, and effective scoring in cognitive evaluation instruments. This work establishes the groundwork for dependable and easily available digital neuropsychological testing by automating the ROCF evaluation process.

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